

LAB REPORT 06

For Loops & Nested Loops in C++

Computer Programming · UET Nowshera · Electrical Engineering

1 Objective

This lab session focused on gaining practical experience with loop constructs in C++. The following objectives were targeted:

- Understanding the concept and syntax of the **for loop** in C++
- Learning how to implement **nested for loops** for structured and repeated operations
- Applying loops to solve problems such as counting, pattern printing, and mathematical computations

2 Theory

2.1 The for Loop

A **for loop** is a control structure that runs a block of code repeatedly as long as a given condition holds true. It is best suited for situations where the number of iterations is known in advance. The loop consists of three parts written inside parentheses: an *initialization* that runs once at the start, a *condition* that is checked before every iteration, and an *update expression* that executes after each pass through the body. The cycle repeats until the condition becomes false.

● ● ● Syntax

```
for (initialization; condition; update) {  
    // body executes while condition is true  
}
```

Execution order: initialization runs once → condition evaluated → if true, body runs → update executes → condition checked again. Continues until condition is false.

2.2 Nested for Loops

A **nested loop** is a loop placed inside another loop. For every single iteration of the outer loop, the inner loop completes its full cycle. This is the standard approach for working with two-dimensional structures such as tables, grids, and printed patterns. The total number of body executions equals the product of both loop counts.

3 Programs Implemented

3.1 — Countdown from 5 to 1

A for loop starts at 5 and decrements until it reaches 0. Each value is printed on its own line, and a message confirms execution continued past the loop.

● ● ● C++ Code

```
#include <iostream>
using namespace std;

int main() {
    for (int i = 5; i > 0; i--) {
        cout << i << endl;
    }
    cout << "outside loop now" << endl;
    return 0;
}
```

3.2 — Prime Number Check

The user enters a number. A for loop tests every integer from 2 up to one less than that number. A flag variable tracks whether a divisor was found, and the result is reported after the loop.

● ● ● C++ Code

```
#include <iostream>
using namespace std;

int main() {
    int num, flag = 0;
    cout << "Enter a number: "; cin >> num;
    for (int i = 2; i < num; i++) {
        if (num % i == 0) { flag = 1; break; }
    }
    if (flag == 1) cout << num << " is not prime" << endl;
    else          cout << num << " is prime" << endl;
    return 0;
}
```

3.3 — Nested Loop Pattern Printing

An outer loop counts down from 7 to 1. For each outer value, an inner loop prints integers 1 through that value, producing a descending staircase pattern.

● ● ● C++ Code

```
#include <iostream>
using namespace std;

int main() {
    for (int num = 7; num > 0; num--) {
        for (int j = 1; j <= num; j++) { cout << j << " "; }
        cout << "\n";
    }
    return 0;
}
```

4

Lab Tasks

4.1 — While Loop: Display 1 to 10

A while loop prints integers 1 through 10, with a counter initialised before the loop and incremented each iteration.

● ● ● C++ Code

```
#include <iostream>
using namespace std;
int main() {
    int i = 1;
    while (i <= 10) { cout << i << endl; i++; }
    return 0;
}
```

4.2 — Multiplication Table Using Nested For Loops

Nested for loops produce a 10x10 multiplication table. The outer loop provides rows and the inner loop columns; each cell prints the product separated by a tab.

● ● ● C++ Code

```
#include <iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 10; i++) {
        for (int j = 1; j <= 10; j++) { cout << i*j << "\t"; }
        cout << endl;
    }
    return 0;
}
```

4.3 — Sum of Numbers from 1 to N

The user enters N. A for loop accumulates the sum of all integers from 1 to N and displays the result.

● ● ● C++ Code

```
#include <iostream>
using namespace std;
int main() {
    int n, sum = 0;
    cout << "Enter a positive integer: "; cin >> n;
    for (int i = 1; i <= n; i++) { sum += i; }
    cout << "Sum = " << sum << endl;
    return 0;
}
```

4.4 — Triangle Pattern Printing

Nested loops print a right-angled triangle of asterisks. The outer loop controls rows; the inner loop prints asterisks up to the current row number.

● ● ● C++ Code

```
#include <iostream>
using namespace std;
int main() {
    int n; cout << "Rows: "; cin >> n;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) { cout << "* "; }
        cout << endl;
    }
    return 0;
}
```

5 Conclusion

This lab gave hands-on experience with for loops, while loops, and nested loops in C++. The tasks demonstrated how loop constructs control program flow across a practical range of applications —

counting, primality testing, multiplication tables, sums, and pattern printing — showing loops to be among the most versatile tools in C++ programming.